Section A - Theory

Choosethecorrectoption(a,b,c, or d) for each question. Write only the letter.

A1. In MySQL, what is the relationship between a schema and a database?

a) Schema is a group of databases
b) Database is a group of schemas
c) Schema and database are the same thing in MySQL
d) Schema is a virtual view of a database

Your answer: C

A2. Which SQL command family does TRUNCATE TABLE belong to?

a) DML (Data Manipulation Language)
b) DDL (Data Definition Language)
c) DQL (Data Query Language)
d) DCL (Data Control Language)

Your answer: B

A3. You are designing a table to store product prices for an e-commerce site (e.g. ₹299.50, ₹79,999.00). Which data type is the most appropriate?

a) FLOAT
b) DECIMAL
c) VARCHAR
d) INT

Your answer: B

A4. You try to INSERT a row into an orders table with a customer_id that doesn't exist in the customers table. There is a foreign key constraint. What happens?

a) The row is inserted with NULL for customer_id
b) The row is inserted and a warning is logged
c) The INSERT fails with a foreign key constraint error
d) The row is inserted but marked as invalid

Your answer: C

A5. Which of the following statements is TRUE about WHERE vs HAVING?

a) WHERE and HAVING can be used interchangeably

b) WHERE filters individual rows before aggregation; HAVING filters groups after aggregation
c) HAVING is used for row-level filtering; WHERE is used for group filtering

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d) HAVING is faster than WHERE

Your answer: __B__

A6. What is the result of the query: SELECT * FROM products WHERE avg_rating = NULL;

a) Returns all rows where avg_rating is NULL

b) Returns all rows where avg_rating is NOT NULL

c) Returns 0 rows (no rows match)

d) Throws a syntax error

Your answer: __C__

A7. What is the difference between COUNT(*) and COUNT(column_name)?

a) They are identical

b) COUNT(*) counts all rows; COUNT(column_name) counts only non-NULL values in that column

c) COUNT(*) is faster but less accurate

d) COUNT(column_name) returns only distinct values

Your answer: __B__

A8. Which command permanently removes an entire table (both structure and data), and CANNOT be undone by ROLLBACK?

a) DELETE FROM table_name

b) TRUNCATE TABLE table_name

c) DROP TABLE table_name

d) REMOVE TABLE table_name

Your answer: __C__

Section B - Output Prediction

Foreachquerybelow,describewhatitreturns.Give the row count and/or specific values.

B1. What does this query return?

```
SELECT COUNT(*) FROM products WHERE category = 'Electronics';
```

Your answer:

This query returns the total number of products in the Electronics category.

B2. How many rows does this query return?

```
SELECT * FROM products WHERE price BETWEEN 1000 AND 3000;
```

Your answer:

11

B3. What is the output of this query?

```
SELECT product_name FROM products
WHERE category = 'Books' AND price < 400
ORDER BY price DESC
LIMIT 1;
```

Your answer:

This query returns the name of the most expensive book that costs less than ₹400. It shows only one product name .

B4. How many rows does this query return?

```
SELECT * FROM products WHERE avg_rating IS NULL;
```

Your answer:

3

B5. What single value does this query return?

```
SELECT MAX(price) FROM products WHERE category = 'Books';
```

Your answer:

This query returns the highest price of the book among the books category .

B6. What does this query return?

```
SELECT category, COUNT(*) FROM products
WHERE is_active = TRUE
GROUP BY category
HAVING COUNT(*) > 4;
```

Your answer:

This query shows each category that has more than 4 active products.

B7. What does the output look like?

```
SELECT product_name,
    CASE
        WHEN price < 500 THEN 'Budget'
        WHEN price < 5000 THEN 'Mid'
        ELSE 'Premium'
    END AS tier
FROM products WHERE category = 'Beauty';
```

Your answer:

product_name	tier
Nykaa Matte Lipstick	Mid
Lakme Eye Liner	Budget
Mamaearth Face Wash	Budget
WOW Skin Vitamin C Serum	Mid

B8. What does this query return?

```
SELECT product_name, COALESCE(avg_rating, 0) AS rating
FROM products
WHERE stock_quantity = 0;
```

Your answer:

This query returns the names and ratings of all products that are out of stock.

Section C - Applied SQL

Writethe actual SQL queryforeachquestion. Formatting counts - use uppercase for keywords and align clauses on new lines for complex queries.

C1 - Basic SELECT + WHERE + ORDER BY

C1. Write a query to display all products with all columns.

Your query:

```
Select *from products;
```

C2. Write a query that shows the product_name and price of all Books.

Your query:

```
Select product_name, price
from products
where category = 'Books';
```

C3. Write a query to list all products priced above ₹10,000, sorted from highest price to lowest.

Your query:

```
select *from products where price>10000 order by price desc;
```

C4. Write a query to display the top 5 most expensive products in the Electronics category (show product name and price).

Your query:

```
select product_name, price from products where category = 'Electronics' order by price desc limit 5;
```

C2 - Filtering Variations (IN, BETWEEN, LIKE)

C5. Write a query to list all products in either Electronics or Apparel category. Use the IN operator.

Your query:

```
select *from products where category = 'Electronics' or category = 'Apparel';
```

C6. Write a query to find all products with a price between ₹500 and ₹2,000 (inclusive).

Your query:

```
select *from products where price between 500 and 2000;
```

C7. Write a query to find all products whose product_name contains the word 'Watch'.

Your query:

```
select *from products where product_name like '%watch%';
```

C8. Write a query to find all products whose brand starts with the letter 'S'.

Your query:

```
select  *from products where brand like 's%';
```

C3 - DISTINCT + Aggregate Functions

C9. Write a query to list all the unique categories in the products table.

Your query:

```
select distinct category from products;
```

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C10. Write a query to find the total number of products in the entire catalogue.
Your query:

```
select count(*) as total_products from products;
```

C11. Write a query to find the average price of all Books.
Your query:

```
select avg(price) as avg_price from products where category='Books';
```

C12. Write a single query that returns the maximum and minimum price across all products.
Your query:

```
select max(price) as max_price,min(price) as min_price from products;
```

C4 - GROUP BY

C13. Write a query to count how many products are in each category.
Your query:

```
select category, count(*) as product_count from products group by category;
```

C14. Write a query to show the total stock quantity for each category.
Your query:

```
select category, sum(stock_quantity) as total_stock from products group by category;
```

C15. Write a query to show the average price per category, sorted from highest average to lowest.

Your query:

```
select category, avg(price) as avg_price from products
group by category order by avg_price desc;
```

C16. Write a query to show the count of products and the average price for each brand, showing only brands that have more than 1 product.

Your query:

```
select brand, count(*) as product_count , avg(price) as avg_price
from products group by brand having count(*)>1;
```

C5 - HAVING + LIMIT

C17. Write a query to find categories that have more than 4 active products.

Your query:

```
select category from products where is_active = 1 group by category having count(*) > 4;
```

C18. Write a query to find the 3 most expensive products in the entire catalogue.

Your query:

```
select product_name, price from products order by price desc limit 3;
```

C19. Write a query to find all categories where the average price is above ₹2,000.

Your query:

```
select category from products group by category having avg(price) > 2000;
```

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C6 - NULL Handling

C20. Write a query to list all products where the avg_rating is missing.

Your query:

```
select * from products where avg_rating is null;
```

C21. Write a query to show product_name and rating for all products. If avg_rating is NULL, display the text 'New Launch' instead.

Your query:

```
select product_name, if null(avg_rating, 'New Launch') as rating from products;
```

C7 - CASE WHEN

C22. Write a query to display each product with a price_tier column classified as 'Budget' if price < ₹1,000, 'Mid' if price < ₹10,000, or 'Premium' if price >= ₹10,000.

Your query:

```
select product_name, price,case
when price < 1000 then 'Budget'
when price < 10000 then 'Mid'
else 'Premium' end as price_tier from products;
```

C23. For each category, write a query that shows the total product count and the count of 'Premium' products (price >= ₹10,000). Use SUM(CASE WHEN...).

Your query:

```
select category , count(*) as total_products ,
sum(case when price >= 10000 then 1
else 0 end) as premium_products from products group by category;
```

C24. Write a comprehensive query showing for each category (only categories with at least 3 products): total product count, count of active products, average price, and a category_tier column ('Cheap' if avg < ₹1500, 'Standard' if avg < ₹10000, 'Luxury' if avg >= ₹10000). Sort by average price descending.

Your query:

```
select category,count(*) as total_products,
sum(is_active) as active_p roducts,
avg(price) as average_price,
case
when avg(price) < 1500 then 'Cheap'
when avg(price) < 10000 then 'Standard'
else 'Luxury' end as category_tier
from products group bycategory
having count(*) >= 3
order by avg(price) desc;
```